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Analysis of formation and growth of the σ phase in additively manufactured functionally graded materials

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ABSTRACT

This study focuses on the formation of the σ phase in three functionally graded material (FGM) systems made by additive manufacturing (AM): stainless steel 420 (SS420) to V to Ti–6Al–4V, Ti–6Al–4V to V to stainless steel 304 L (SS304L), and SS420 to V. Directly joining Ti–6Al–4V and stainless steel may result in the formation of brittle Fe–Ti intermetallics. This study investigates the potential use of V as an intermediate element between terminal alloys of Ti–6Al–4V and stainless steel. Experimental analysis of the elemental and phase composition revealed that different amounts of σ phase were present in the three FGM systems at locations with similar elemental compositions. Computational studies were performed to simulate the thermal history, phase transformation kinetics, and σ phase growth within these FGMs. The computations suggested that, at the conditions studied, the σ phase should nucleate faster and grow to a larger volume fraction in the SS420–V alloy than the SS304L–V alloy, contrasting with experimental observations. Instead, experimental analysis confirmed that the disparate growth of σ phase in the FGMs was due differences in cracking during fabrication, resulting in different amounts of time spent at the elevated temperatures conducive to σ phase growth in each of the samples.

1. Introduction

The properties of functionally graded materials (FGMs) vary over one or more dimension due to intentional changes in chemistry or microstructure within a single component. Metallic functionally graded materials can be fabricated using directed energy deposition (DED) additive manufacturing (AM) [1–3]. In this layer-by-layer fabrication method, a laser beam creates a melt pool on the substrate, or subsequent layers, and powder feedstock is deposited into the melt pool, which solidifies and fuses to the substrate or layer below. This process is repeated to create a 3D component. During the DED process, the relative volume fraction of different powders can be changed during processing, allowing for the deposition of different compositions as a function of position within a 3D component.

In FGMs, the length scales over which the chemistry is changed can be varied. Small incremental changes, on the order of a 1–3 vol% per layer, result in smooth transitions in composition over lengths of tens of millimeters [3,4]. In this study, we investigated FGMs that had jumps in composition, ranging from 25 to 100 vol%, between subsequent sections within each FGM, allowing for the investigation of re-melting and mixing present in adjacent layers, with drastically different chemistries, in AM. One combination of dissimilar materials that is of interest to industries such as aerospace [5,6] and nuclear [7–9] is Ti–6Al–4V and stainless steel [7,10]. Combining these alloys using traditional joining techniques, such diffusion bonding, fusion welding, and friction stir welding, results in weak joints between the two materials [6,11]. The main cause of the weakness in welded components is the formation of brittle Fe–Ti intermetallic phases, such as FeTi and Fe_2Ti [11–13]. Ti–6Al–4V has been successfully joined to micro-duplex stainless steel using diffusion bonding, but this process is limited by the

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geometry of the sample, which restricts the applications for which it can be used [7]. Additive manufacturing has been explored as a means of joining Ti-alloys and stainless steel with limited success, as intermetallics form at the interface between the two materials in a bimetallic structure [14].

There have been several studies that explore the addition of intermediate elements between Ti–6Al–4V and stainless steel to join these terminal alloys. For example, pure V has been introduced as an interlayer between Ti–6Al–4V and 316 L stainless steel using laser welding [15], which resulted in the absence of Fe–Ti intermetallic phases. Compositional analysis indicated that the part of the weld where the V interlayer was joined with 316 L stainless steel accessed the composition space of σ-FeV phase based on the Fe–Cr–V isothermal equilibrium phase diagram (at 480 °C), but no evidence of this phase was found through X-ray diffraction. The authors postulate that the high cooling rates during processing resulted in the formation of a (Fe,V) solid solution rather than the σ-FeV phase.

Due to the high cooling rates associated with the process, the additive manufacturing of FGMs of Ti-alloys to stainless steels with intermediate elements or alloys as buffers between the terminal alloys has been explored as a possible means for joining these two systems. A prior study on an FGM made by DED AM and graded from Ti–6Al–4V to 304 L stainless steel used V as an intermediate metal between the terminal alloys [17]. The FGM was graded from 100 vol% Ti–6Al–4V to 25 vol% Ti–6Al–4V/75 vol% V in 25 vol% increments. The gradient sample transitioned from a 25 vol% Ti–6Al–4V/75 vol% V to a 25 vol% 304 L stainless steel/75 vol% V composition, and at this transition the sample cracked during fabrication. X-ray diffraction analysis of the crack surface indicated the presence of intermetallic FeTi. Li et al. examined an FGM fabricated through DED AM that used intermediate layers of 100 vol % V, 100 vol% Cr, and 100 vol% Fe to grade from 100 vol% Ti–6Al–4V to 100 vol% 316 stainless steel [18]. While the number of layers and layer heights for these intermediate elements were not specified, the EDS point scan data indicated re-melting and mixing between the V and Cr layers, with minimal mixing between the other elements/alloys in the gradient region. The incorporation of these single-element layers between Ti–6Al–4V and SS316 did not result in the formation of Fe–Ti intermetallic phases in the sample. Given that the Fe and Cr layers were adjacent to each other, their mixing could have resulted in the formation of the σ-FeCr phase, but this phase was absent due to the fast cooling rates (greater than on the order of 0.1 °C/s [19]) during fabrication [18].

In a study where a V interlayer was introduced, either through diffusion bonding or selective laser melting, between a Ti–6Al–4V baseplate and 17-4 PH stainless steel deposited by DED, the σ-FeV phase was not seen in either as-fabricated sample [16]. However, with post-processing heat treatment, the σ-FeV phase precipitated at the interface between the V interlayer and 17-4 PH stainless steel in the diffusion bonded sample, with the amount of σ-FeV phase increasing with increased time at high temperatures, highlighting the role of prolonged heating on the nucleation and growth of the σ-FeV phase.

Relying solely on cooling rates as a method of mitigating deleterious phase formation in AM is not sufficient, due to the fact that the cooling rate can depend on the component shape and wall thickness, which can then influence the volume fractions of detrimental phases within the sample. Therefore, determining detrimental phase formation calls for a thermodynamic approach rather than a kinetic approach.

The goal of the present work was to determine what led to disparate formation of the σ phase in three FGM systems with similar compositions: a previously analyzed Ti–6Al–4V to V to SS304L FGM (Ti–6Al–4V/V/SS304L) [20], a SS420 to V FGM (SS420/V), and a SS420 to V to Ti–6Al–4V FGM (SS420/V/Ti–6Al–4V). These gradients were chosen due to the similarities in the compositions of SS304L and SS420 and the tendency of both alloys to form the σ phase when joined with V, as evidenced by the Fe–Cr–V ternary phase diagram in Fig. 1. The σ phase, a brittle intermetallic phase that can result in loss of ductility, the formation of cracks, and the reduction of the mechanical integrity of components [21], has been experimentally found in Fe–Cr and Fe–V systems over a range of compositions and temperatures [22]. The experimental characterization in this study focused on the phase and elemental composition of the three FGMs to measure and compare the differences in amounts of the σ-Fe-(Cr,V) phase. The σ-Fe-(Cr,V) phase, henceforth referred to as σ phase, is a low temperature phase that forms from the high temperature bcc phase through a solid state reaction. The presence of any σ phase is therefore a result of a solid-state nucleation and growth process. Time-temperature-transformation curves were simulated using TC-PRISMA to compare the relative nucleation rates of the σ phase in the SS304L–V bcc matrix and SS420–V bcc matrix. Isothermal diffusion-controlled growth of nucleated σ phase was simulated using the DIffusion Controlled TRAnsformations (DICTRA) module of Thermo-Calc and compared for the Ti–6Al–4V/V/SS304L and SS420/V/Ti–6Al–4V FGM systems. Finite element analysis (FEA) was performed to simulate the thermal history in the Ti–6Al–4V/V/SS304L and SS420/V/Ti–6Al–4V FGMs in order to determine an approximate initial cooling rate, as well as the subsequent heating and cooling cycles in the AM fabrication process, which could contribute to the σ phase growth. The present work highlights the importance of using a combination of computational and experimental methods to fully understand the growth and nucleation of the σ phase in additively manufactured functionally graded systems.

2. Experimental methods

All three samples were fabricated using a directed energy deposition additive manufacturing system (RPM 557 Laser Deposition System), shown schematically in Fig. 2. Deposition for all three FGMs was performed with a laser scanning speed of 12.7 mm/s and a hatch spacing of 0.58 mm in an Ar atmosphere to prevent oxidation. The layer height was 0.3 mm for the SS420/V/Ti–6Al–4V FGM and 0.38 mm for the Ti–6Al–4V/V/SS304L and SS420/V FGMs.

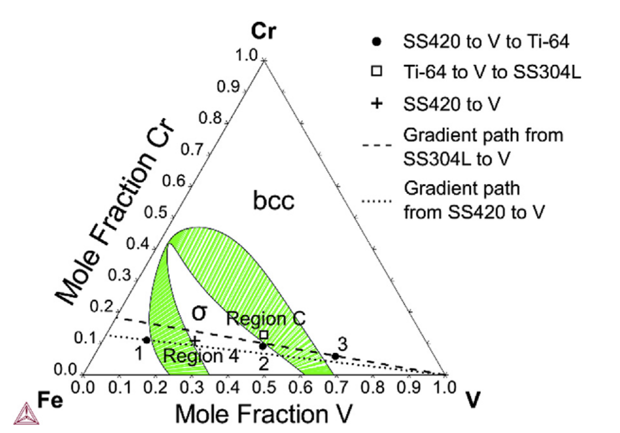

Fig. 1. Ternary isothermal (at 850 °C) section of the Fe–Cr–V system calculated using the TCFE8 database [31]. The compositions noted by the various symbols correspond to Region C in the Ti–6Al–4V/V/SS304L FGM (Fig. 5), Region 4 in the SS420/V FGM (Fig. 4), and Compositions 1–3 from Fig. 7a of the SS420/V/Ti–6Al–4V FGM. The lines correspond to the direct gradient path between SS304L and V (dashed line) and SS420 and V (dotted line).

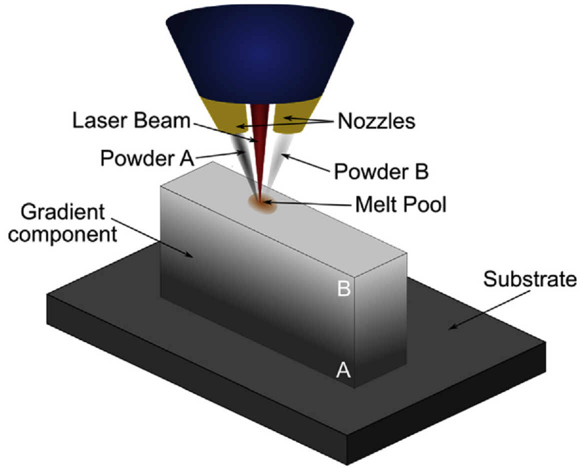

Fig. 2. Schematic of the DED process used for fabricating an FGM.

The hatch angle between subsequent layers varied in the three FGMs: the hatch angle was 90° in the SS420/V/Ti–6Al–4V FGM, 30° in Ti–6Al–4V/V/SS304L FGM, and 45° in the SS420/V FGM. The DED system used a YAG laser operated at varying laser powers and laser spot sizes for each FGM studied. Table 1 provides the compositions of each FGM and the laser parameters used to fabricate each sample. The laser power was altered for different compositions in the SS420/V and SS420/V/Ti–6Al–4V FGMs in order to account for the differences in thermal properties of SS420, V, and Ti–6Al-4V, as discussed in Section 4.2 and in Ref. [13]. The powder diameter ranges used were 44 μm–177 μm (−80/+325 mesh size) for the pre-alloyed Ti–6Al–4V, 45 μm–250 μm (−60/+325 mesh size) for V, and 44 μm–160 μm (−140/+325 mesh size) for pre-alloyed SS420. The compositions of all of the powders are given in Table 2.

For the SS420/V/Ti–6Al–4V FGM, 25 layers of 100 vol% SS420 were deposited, followed by 2 layers of 100 vol% V, and ending with 25 layers of 100 vol% Ti–6Al–4V. The final height of the sample was approximately 15.6 mm. A cross-sectional macrograph of the sample is shown in Fig. 3a. The composition of the SS420/V FGM was graded in 10 vol% increments every 20 layers, starting at 100 vol% SS420. These gradient compositions are labeled as Regions 1–4 in Fig. 4a, a macrograph of the cross-section of the sample. The intent was to grade to 100 vol% V; however, due to cracking in Region 4, the build was terminated at a composition of 70 vol% SS420/30 vol% V. A detailed description of the fabrication process of the Ti–6Al–4V to V to SS304L FGM is given in Ref. [20], and the relevant compositions and corresponding processing parameters are given in Table 1. A macrograph of the cross-section of the final three gradient composition regions, labeled Regions A, B, and C, is given in Fig. 5a.

The Ti–6Al–4V/V/SS304L and SS420/V/Ti–6Al–4V FGMs were removed from their respective baseplates, and all three FGMs were cut vertically using wire electrical discharge machining (EDM). One half of each sample was mounted and polished using standard metallographic techniques with a final polish using a 0.05 μm silica suspension. The samples were then etched using Kroll's reagent (2 vol% hydrofluoric acid and 3 vol% nitric acid diluted in water).

Energy dispersive X-ray spectroscopy (EDS) was used to perform elemental analysis through line scans and composition maps of the SS420/V/Ti–6Al–4V FGM. Point composition analysis was performed in selected areas of the SS420/V and Ti–6Al–4V/V/SS304L FGMs. EDS analysis was performed using a scanning electron microscope (SEM, FEI Quanta 200 and Quanta 3D FEG Dual Beam) with a silicon drift detector (Oxford X-Max and Oxford X-act PentaFET Precision). X-ray diffraction (XRD) and electron back scatter diffraction (EBSD) were used for phase identification. XRD patterns of the entire transition region of the SS420/V/Ti–6Al–4V FGM were collected using a Bragg-Brentano-type diffractometer (Panalytical Empyrean, Cu K-α X-ray source, 45 kV, 40 mA, $\lambda = 1.54$ Å). EBSD (Oxford NordlysNano and Oxford Nordlys Max2) was performed on localized regions of the transition zone of the SS420/V/Ti–6Al–4V FGM, within Region C of the Ti–6Al–4V/V/SS304L FGM, and within Region 4 of the SS420/V FGM.

3. Computational simulations

The finite element software, Netfabb Simulation [23], was used to approximately simulate the thermal history during fabrication at the location where σ phase was found or expected to form in the SS420/V/Ti–6Al–4V and Ti–6Al–4V/V/SS304L FGMs. A simplified geometry for both systems was considered, consisting of a substrate whose height was equal to the height of the actual substrate plus the height of the sample up to the point of interest. The rectangular surface area of the model substrate was equal to the footprint of the actual sample, and the layers were deposited as a single track along the length of the substrate, with subsequent layers stacked on top of this single track. In Netfabb, the laser path is explicitly considered, with the laser power, scan speed, layer height, and laser absorption efficiency specified. Additionally, the temperature-dependent thermal conductivity and specific heat, density, and latent heat of the material, were considered. The material properties and processing parameters used in these simulations are given in Table 3. The thermal history during fabrication was extracted between the last layer of SS420 and the first layer of V in the SS420/V/Ti–6Al–4V FGM and between the last layer of the

Table 1
Information on the fabrication of the FGMs, including the number of layers for each composition and the laser parameters used at those compositions.

FGM	Number of Layers	Vol. % SS304L	Vol. % SS420	Vol. % V	Vol % Ti–6Al–4V	Laser Power (W)	Laser Spot Size (mm)
SS420/V/Ti–6Al–4V	25	–	100	0	0	600	1.02
	2	–	0	100	0	800	1.02
	25	–	0	0	100	450	1.02
Ti–6Al–4V/V/SS304L	28	0	–	0	100	600	1.02
	27	0	–	25	75	600	1.02
	27	0	–	50	50	600	1.02
	27	0	–	75	25	600	1.02
	27	25	–	75	0	600	1.02
	27	50	–	50	0	600	1.02
SS420/V	20	–	100	0	–	800	1.40
	20	–	90	10	–	860	1.46
	20	–	80	20	–	920	1.52
	20	–	70	30	–	980	1.59

Table 2
Composition (in wt%) of SS420, V, Ti–6Al–4V, and SS304L powders used for fabrication of the three FGMs.

	C	Mn	P	S	Si	Cr	Ni	Fe	Al	V	O	Ti
SS420	0.26	0.31	0.019	0.004	0.45	13.30	–	Bal.	–	–	–	–
V	0.05	–	–	–	0.002	0.008	–	0.02	0.008	Bal.	–	–
Ti–6Al–4V	0.03	–	–	–	0.08	–	–	0.18	6.21	4.05	0.18	Bal.
SS304L	0.015	1.5	0.012	0.003	0.53	18.4	9.8	Bal.	–	–	–	–

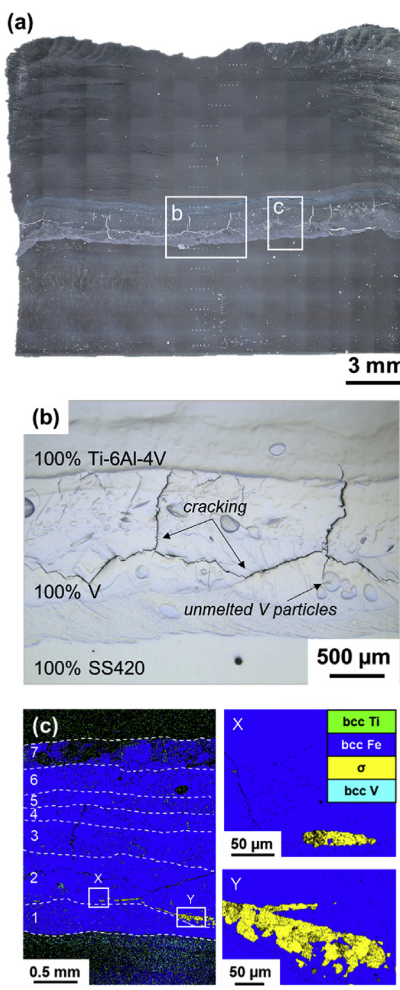

Fig. 3. (a) Optical macrograph of the cross-section of the SS420/V/Ti–6Al–4V FGM with inset (b) highlighting the transition regions between SS420, V, and Ti–6Al–4V. (c) EBSD phase maps of the transition region with inset images X and Y showing the localization of σ phase formation.

Fig. 4. Optical micrograph of the cross-section of the SS420/V FGM, with the composition changes marked and labeled Regions 1–4.

25 vol% SS304L/75 vol% V composition and the first layer of the 50 vol% SS304L/50 vol% V composition in the Ti–6Al–4V/V/SS304L FGM.

Two other types of simulations were performed to compare the cooling rates in FGMs. Finite element analysis simulations that directly modeled the sample geometry, but that were limited in that the laser path and heat input could not be directly modeled, were performed. A simulation method that considers the heat transfer and fluid flow within the melt pool was also performed, and while this method takes into account the molten pool geometry and loss of heat due to convection and radiation, it only considers a single melt pool track and cannot accurately capture the larger geometry and deposition of additional layers, which impact the thermal history profile of the FGMs. For these reasons, and the advantages of Netfabb outlined above, the results from the Netfabb simulation are the only ones reported here.

Thermodynamic analysis of the system was performed to predict the equilibrium phase relations of the σ-containing regions in the Ti–6Al–4V/V/SS304L FGM and the SS420/V/Ti–6Al–4V FGM. These calculations were performed using the CALculation of PHAse Diagrams (CALPHAD) method [24–28], which uses the compound energy formalism [29] to parameterize the Gibbs energy of individual phases with multicomponent sublattice models and corresponding elemental diffusion coefficients. Thermodynamic calculations and kinetic simulations of quaternary alloys of Fe-40.9V-11.8Cr-6.8Ni (wt. %) and Fe-39.6V-9.0Cr-1.9Ti (wt. %) representing the σ-containing regions in the Ti–6Al–4V/V/SS304L and SS420/V/Ti–6Al–4V FGMs, respectively, were performed. These compositions will be referred to as the SS304L–V and SS420–V alloys when discussing the thermodynamic and kinetic calculations.

Time-temperature-transformation (TTT) curves for the nucleation of the σ phase in solid bcc SS304L–V and SS420–V alloys were calculated using the TC-PRISMA module of Thermo-Calc. TC-PRISMA is based on the Kampmann and Wagner Numerical (KWN) model [30], which considers the simultaneous nucleation, growth, and coarsening of precipitate phases in multicomponent alloys. The KWN model requires a description of the driving forces to nucleate the precipitate phase from the supersaturated solution (further referred to as the solution), molar volumes of the solution and precipitate phases, diffusivity in the solution, and the interfacial energy between the solution and precipitate. In TC-PRISMA, driving forces and molar volumes are calculated using the thermodynamic description from the TCFE8 database [31]. Diffusivities of components in each phase are calculated by using the MOBFE3 atomic mobility database [32] coupled with TCFE8 for the thermodynamic factors required to calculate diffusion coefficients. Interfacial energies have not been comprehensively studied and are

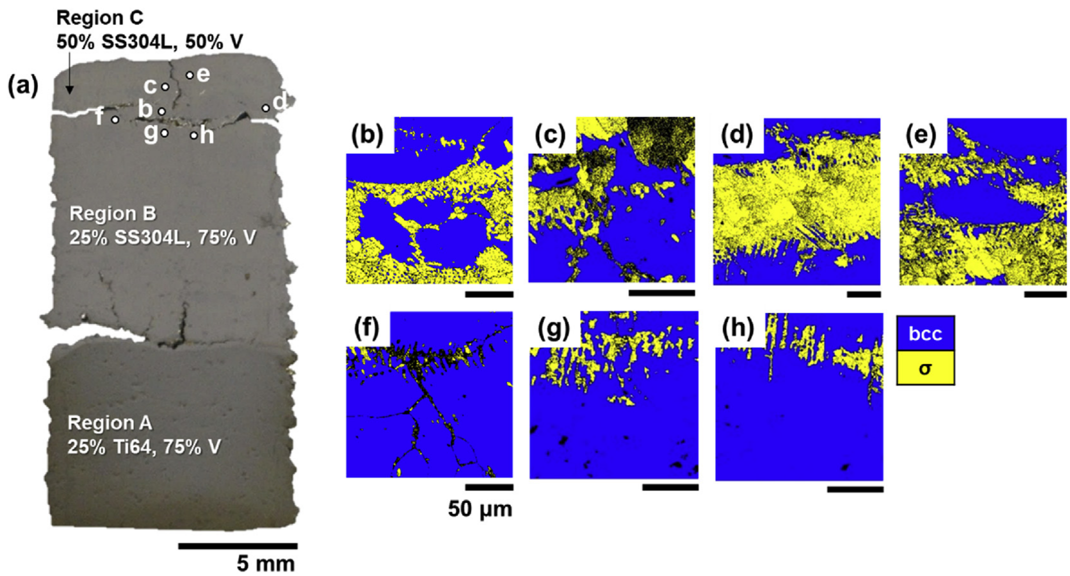

Fig. 5. (a) Cross-sectional macrograph of the Ti–6Al–4V/V/SS304L FGM with composition changes marked and labeled Regions A, B, and C on the macrograph . White dots within Region C indicate the locations of the EBSD phase maps from (b)–(e) areas above the crack that show significant amounts of σ phase and (f)–(h) areas below the crack that show that the majority phase in this region is the bcc phase. Scale bar in (b)–(g) is 50 μm.

Table 3
Materials properties and processing parameters used in the Netfabb simulations.

	Ti–6Al–4V/V/SS304L FGM	SS420/V/Ti–6Al–4V FGM
Thermal conductivity (10^{-3} W/mm-°C, °C)	26.06, 25 35.03, 700 44.75, 1910 [37–39]	23.95, 22 24,90, 100 25.98, 900 [37]
Specific heat (J/kg-°C, °C)	497, 27 576, 127 633, 277 698, 627 870, 1427 [37,39]	331, 22 460, 100 569, 900 [37]
Density (kg/mm^3)	5.5 [37]	7.74 [37]
Latent heat (10^3 J/kg, °C)	381 [37]	411 [37]
Emissivity	0.29 [40]	0.4 [41]
Power (W)	600	800
Laser velocity (mm/s)	12.7	12.7
Laser spot size (mm)	1.016	1.016

computationally challenging to determine, so the approximate interfacial energies were calculated automatically using TC-PRISMA to treat the different alloys similarly.

The relative growth rates of the σ phase in the SS304L–V and SS420–V alloys were investigated using the DICTRA module of Thermo-Calc. DICTRA uses a combination of a CALPHAD-based description of the Gibbs energy of and atomic mobility in individual phases to simulate the one-dimensional moving boundary problem assuming local equilibrium at the interface. The model considers the geometry shown in Fig. 6a with a small amount of the σ phase precipitated in a bcc matrix with the overall composition matching the nominal alloy composition. Isothermal simulations at 1100 K considering this geometry, but with different chemistries corresponding to the SS420–V and SS304L–V alloys, were used to investigate the different relative growth rates for the case of σ phase in these alloys. The TCFE8 database and MOBFE3 database were used for the thermodynamic and atomic mobility descriptions, respectively, of the bcc and σ phases. The mobility of the σ phase is not assessed in the MOBFE3 database; however, the diffusivity of Cr, Fe, Ni, and V in σ are expected to be much lower than in the bcc matrix, so the σ phase can be reasonably approximated as non-diffusing.

4. Results

4.1. Experimental results

The following sections detail the results from experimental characterization of the elemental and phase compositions of the three FGM samples: SS420/V/Ti–6Al–4V, Ti–6Al–4V/V/SS304L, and SS420/V.

4.1.1. SS420 to V to Ti–6Al–4V

In the SS420/V/Ti–6Al–4V FGM, the transitions from 100 vol% SS420 to 100 vol% V and from 100 vol% V to 100 vol% Ti–6Al–4V were analyzed to determine the effect of using V as an intermediate element to grade between the two terminal alloys. The changes in composition are shown in Fig. 3b and the entire region will be referred to as the transition region. In the 100 vol% V region, large horizontal cracks spanned approximately 10 mm of the 17.7 mm width of the sample, with smaller (~0.7 mm) vertical cracks extending through the two V layers and into the first Ti–6Al–4V layer. Un-melted V powder particles were also present throughout the V layers.

EDS line scans spanning vertically through the FGM were used to determine the chemistry as a function of location. Fig. 7a shows the planned and experimentally measured compositions of the three main constituent elements, Fe, V, and Ti. This graph shows that the measured composition deviates from the idealized stepped composition. In the transitions from SS420 to V to Ti–6Al–4V, there are seven distinct composition plateaus, despite there being only two prescribed deposition chemistry changes. These composition plateaus are numbered in Fig. 7, and will be referred to as Compositions 1–7. The EDS composition maps in Fig. 7b provide a qualitative assessment of the distinct changes in composition throughout the transition region, both as a result of the planned changes in composition, as well as re-melting and mixing between layers. At the transition between the last layer of the deposited

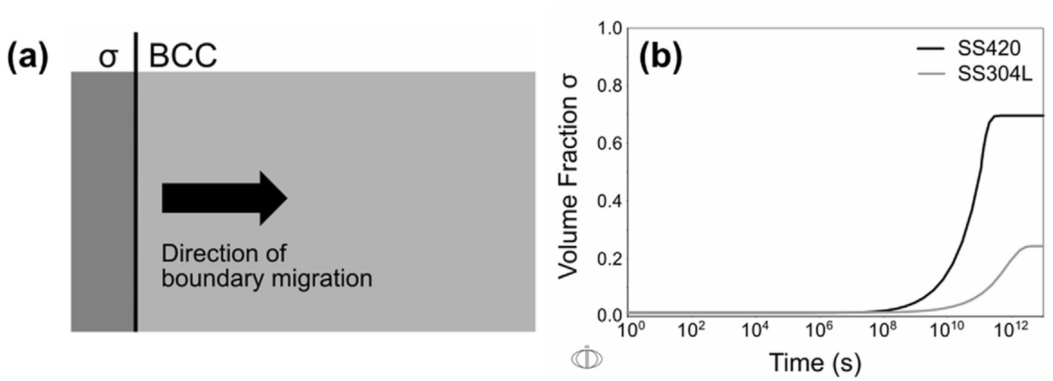

Fig. 6. (a) Representative schematic of the geometry used in DICTRA simulations of the growth of the σ phase in both the SS420–V and SS304L–V alloys. The σ phase has the composition corresponding to the constitution of σ phase in the two phase σ/bcc region. The bcc region has a composition near the nominal composition, that has been mass balanced to the overall composition. The arrow indicates the growth direction of the σ phase, which will grow at the expense of the bcc phase. (b) Time evolution of the DICTRA simulated volume fraction of the σ phase in the 50 nm box at 1100 K for the SS420–V alloy (black line) and SS304 L–V alloy (gray line).

100 vol% SS420 gradient region and the first layer of the planned 100 vol% V gradient region, intermixing is evidenced by V in the planned 100 vol% SS420 gradient region and Fe in the planned 100 vol% V gradient region. At Compositions 1 and 2, the intended composition was 100 vol% SS420, but the measured composition shows that there is between 10 wt% and 50 wt% of V present, respectively. Similarly, Composition 3 corresponds to the prescribed 100 vol% V deposition, but 35 wt% of Fe is present. Finally, the chemistry of Composition 4 (55 wt% Ti, 35 wt% V, 10 wt% Fe) is indicative of the re-melting between the second layer of the V deposition and the first layer of the Ti–6Al–4V deposition.

The diffraction pattern from XRD analysis of the transition

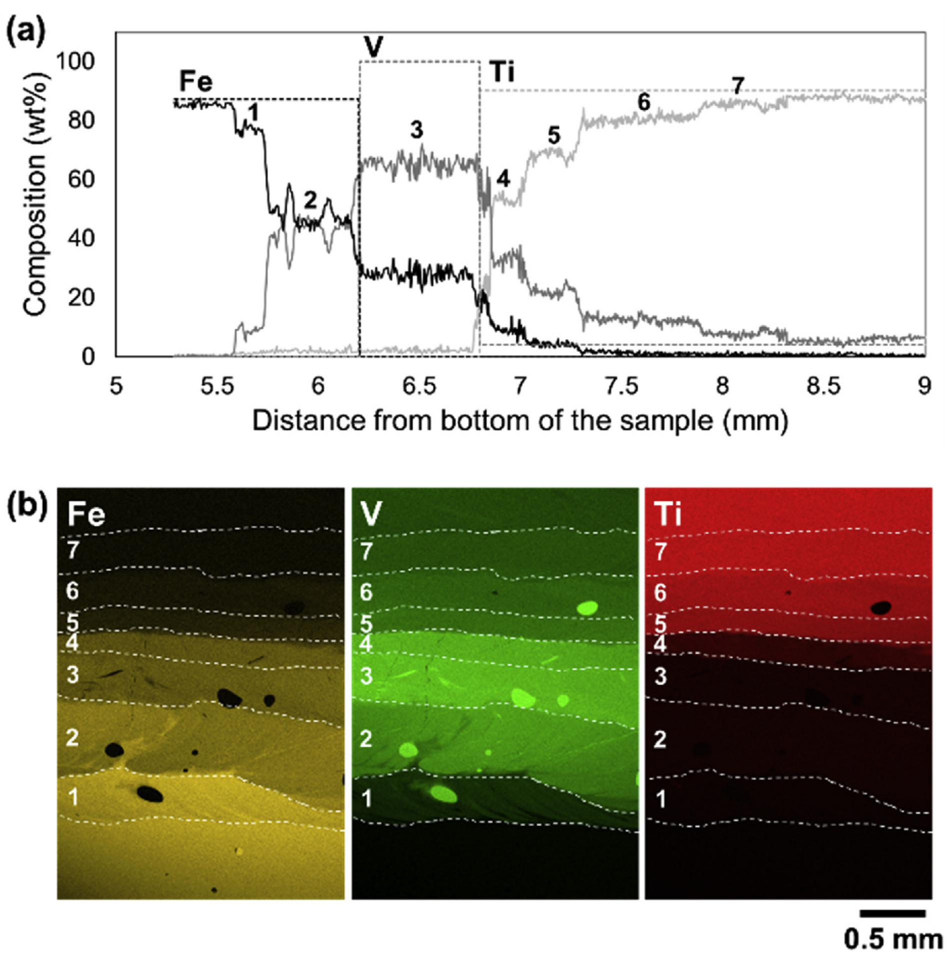

Fig. 7. (a) Composition within the transition region in the SS420/V/Ti–6Al–4V FGM. The dashed lines represent the planned composition while the solid lines represent the measured composition (via EDS). The numbers correspond to seven distinct chemistry plateaus within the gradient region of the FGM. (b) EDS composition maps of the three main constituent elements, Fe, V, and Ti where the numbers correspond to the distinct chemistry plateaus observed in the EDS line scans shown in (a).

region, spanning a 6 mm × 10 mm area, which includes the last few layers of deposited SS420, the entire V region, and the first few layers of deposited Ti–6Al–4V, is shown in Fig. 8. This contains peaks corresponding to the bcc phase, α-Ti (hcp) and β-Ti (bcc) phases, and σ phase. Localized phase analysis was performed using EBSD and the resulting maps are given in Fig. 3c, with phase fraction data given in Table 4. The phase map in Fig. 3c spans the transition region from SS420 to V to Ti–6Al–4V. The lack of σ phase in this presented area of the SS420/V/Ti–6Al–4V FGM is representative of the entire transition region within this FGM where, although the chemistry should result in the equilibrium formation of σ phase, the σ phase was scarce or absent. The primary phase throughout the transition region is the bcc phase. While Fe and V are both bcc, Ti–6Al–4V has both hcp and bcc phases, with the hcp phase being dominant at room temperature; however, V is a bcc stabilizer; thus, adding V to Ti–6Al–4V results in a primarily bcc composition [33]. The insets X and Y in Fig. 3c highlight the small volume fraction of the σ phase (0.41 area%) present in composition 2. The elemental compositions for the insets were measured using EDS; the composition of inset X is 47 wt% Fe, 45 wt% V, and 8 wt% Cr and for inset Y it is 46 wt% Fe, 47 wt% V, and 7 wt% Cr.

4.1.2. Ti–6Al–4V to V to SS304L FGM

Extensive analysis of the Ti–6Al–4V/V/SS304L FGM is presented in Refs. [17,20]. Prior studies focused on the chemistry, phase composition, and microhardness in Regions A, B, and C [34]. The hardness was highest in Region C, corresponding to the presence of a notable amount of σ phase [34]. In the present study, the focus is on this region, which is 50 vol% SS304L/50 vol% V (Region C in Fig. 5). EBSD was performed on areas above and below the large crack in Region C as shown in Fig. 5, with map locations superimposed on the image of the cross-section of the sample. Fig. 5 includes four phase maps from above the crack and three phase maps from below the crack, all within Region C. The average phase fraction data for the phase maps above and below the crack are given in Table 4. Above the crack, 5.9% of the area could not be indexed and below the crack, 8.1% of the total area could not be indexed and these areas are referred to as the zero solution. The EBSD phase maps for these areas are given in Fig. 5b-e and the average measured phase composition from these areas show that 33.1% of the total area is σ phase and 61.0% of the total area is bcc. In contrast, the area below the crack (Fig. 5f-h) has an average phase composition of 2.5 area% σ phase and 89.4 area% bcc.

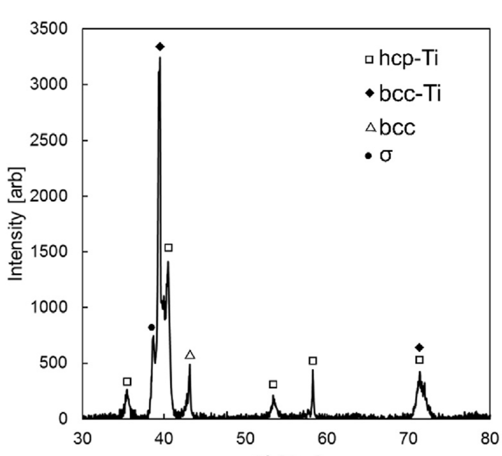

Fig. 8. X-ray diffraction pattern of the transition zone of the SS420/V/Ti–6Al–4V FGM showing the presence of bcc, hcp/bcc-Ti, and σ-FeV.

4.1.3. SS420 to V FGM

In the SS420/V FGM, the experimental analysis focused on the 70% SS420/30% V gradient composition region, referred to as Region 4 in Fig. 4. EBSD analysis was performed on the areas surrounding the crack in Region 4 as shown in Fig. 9b, c, d, and e, with the locations of data acquisition noted on the optical micrograph in Fig. 9a. These phase maps span the entirety of Region 4. The phase composition surrounding the crack on the right side was similar to that surrounding the crack on the left. The phase fraction data for these four phase maps are given in Table 4. Directly above the crack (Fig. 9b) there is a significant amount (62.9 area%) of the σ phase, with 16.1 area% of α-Fe phase, and 14.7 area% zero solution. Directly below the crack (Fig. 9c) the primary phase is the α-Fe phase (76.5 area%) with small amounts of σ phase (12.4 area%). The analyzed areas from above (Fig. 9b) and below (Fig. 9c) the crack were from the same gradient composition region, indicating that the difference in phase composition is not derived from a difference in composition.

In order to determine if the location of an area, in relation to the distance from the crack, impacts the amount of σ phase present, EBSD phase maps were acquired directly adjacent to the crack and from a region in the center of the sample, far away from either of the cracks that formed on the outer edge of the sample. These phase maps are given in Fig. 9d and e, and the phase fraction data are given in Table 4. Fig. 9d shows that directly adjacent to the crack the σ phase accounts for only 16.4 area%, while the α-Fe phase is the majority phase (50.8 area%). In the center of the sample, the EBSD phase map (Fig. 9e) indicates that the α-Fe phase accounts for a majority of the overall area (71.2 area%) and there is almost no σ phase (3.0 area%).

The elemental composition of the regions identified as α-Fe and σ phase were measured using EDS. Tables 5 and 6 give the elemental compositions of the overall analyzed areas of Fig. 9b and c, respectively, as well as the compositions of the α-Fe and σ phases. The measured elemental composition of the overall area for both maps is comparable to the expected deposited composition (65.0 wt% Fe, 10.6 wt% Cr, 24.4 wt%). The measured elemental compositions of the σ and α-Fe phases are similar to each other and to the composition of the overall area. This confirms that there is no elemental compositional difference that led to an increase in the amount of σ phase in the region above the crack.

In addition to the EBSD phase maps, inverse pole figure (IPF) maps of the same areas were also acquired, as shown in Fig. 9b-e. These maps were used to compare the in-plane areas of the grains, referred to as the grain size, for the σ phase and the α-Fe phase as summarized in Table 7. The key difference between these locations is the difference in size of the σ phase grains in the area directly above the crack compared to the areas below and adjacent to the crack, as well as in the center of the sample. In the area directly above the crack, the average grain size of the σ phase was about three times larger than that of the grains in the area below and adjacent to the crack and about six times larger than that of the grains in the center of the sample, far away from cracking on either side of the sample. This difference in grain size of the σ phase indicates that grain growth in an area was affected by the proximity of that area to the cracks along the outer edge of the sample.

4.2. Computational results

Thermodynamic calculations were performed to analyze the phase relations in the transition zone of the FGMs. The Fe–Cr–V ternary phase diagram was used in conjunction with the experimental elemental composition results to analyze the phase composition of the three FGM systems. The isothermal section at 850 °C for the Fe–Cr–V system, as assessed in TCFE8, is shown in

Table 4
EBSD phase fraction data for the SS420 to V to Ti−6Al−4V, Ti−6Al−4V to V to SS304L, and SS420 to V FGMs. These data correspond to the EBSD phase maps given Figs. 3, 5 and 9, respectively. The phase fraction data for the Ti64/V/SS304L FGM is an average of the four images from above the crack and the three images below the crack. Phases not included in a specific system are marked as not applicable (n/a).

	SS420/V/Ti−6Al−4V	Ti−6Al−4V/V/SS304L		SS420/V			
		Above	Below	Above	Below	Adjacent	Center
σ	0.4%	33.1%	2.5%	62.9%	12.4%	16.4%	3.0%
bcc	66.8%	61.0%	89.4%	n/a	n/a	n/a	n/a
α-Fe	n/a	n/a	n/a	14.7%	76.5%	50.8%	71.2%
Ti	3.4%	n/a	n/a	n/a	n/a	n/a	n/a
Zero solution	29.3%	5.9%	8.1%	22.4%	11.1%	32.8%	25.8%

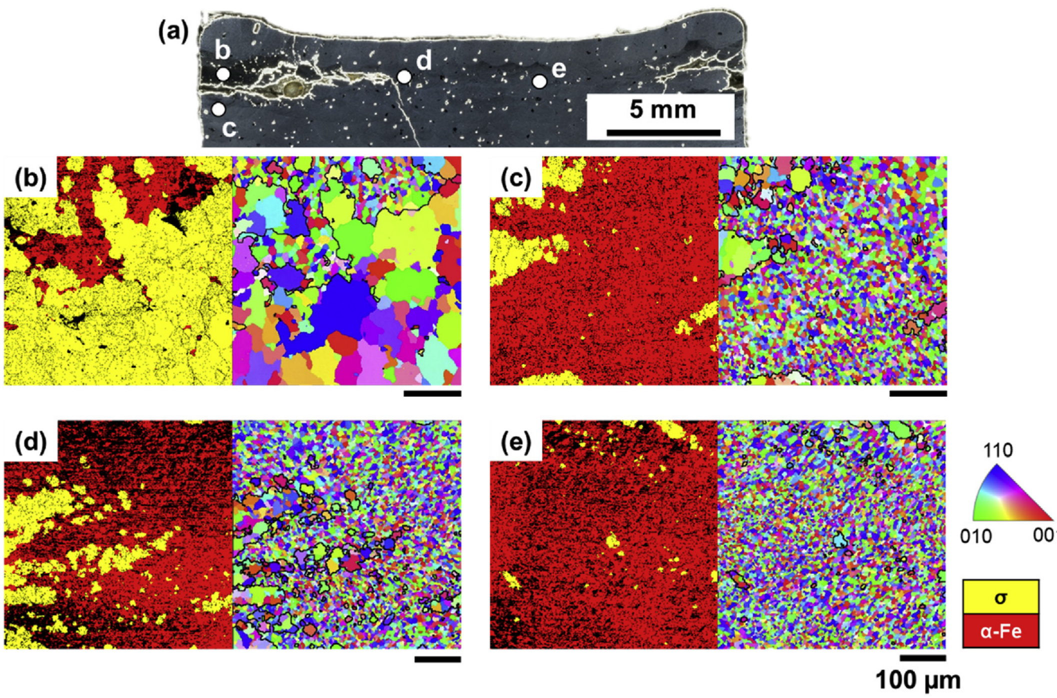

Fig. 9. (a) Optical macrograph of the top of the SS420/V FGM showing the location of the EBSD phase maps and IPF maps. The EBSD phase maps show the amounts of the σ and α-Fe phases present in certain areas, while the IPF maps highlight the difference in the grain size between these two phases. (b) Area above the crack, (c) area below the crack, (d) area adjacent to the crack, and (e) area in the center of the sample. The dark black lines on the IPF maps indicate the phase boundaries. All scale bars in (b)–(e) are equal to 100 μm.

Table 5
Elemental composition of the overall area, σ phase, and α-Fe phase of Fig. 9b.

	Overall area	σ	α-Fe
Wt% Fe	63.8	63.7	63.1
Wt% Cr	10.8	10.9	10.8
Wt% V	25.4	25.4	26.1

Table 6
Elemental composition of the overall area, σ phase, and α-Fe phase of Fig. 9c.

	Overall area	σ	α-Fe
Wt% Fe	65.3	63.4	66.0
Wt% Cr	11.1	10.9	11.2
Wt% V	23.6	25.7	22.8

Fig. 1. Compositions 1−3 of the SS420/V/Ti−6Al−4V FGM, as identified in the EDS line scan data in Fig. 7a, are marked on the phase diagram. Compositions 4−7 are not plotted, as Ti is the majority component and Ti is not considered in this phase diagram due to the errors in the Fe−Ti binary system in the TCFE8 as discussed

Table 7
Two-dimensional analysis of σ phase and α-Fe phase grains from the inverse pole figures given in Fig. 9.

	Above crack		Below crack		Center of sample		Right of crack	
	σ	α-Fe	σ	α-Fe	σ	α-Fe	σ	α-Fe
Average (μm^2)	600	71	174	42	97	48	160	53
Max (μm^2)	9225	556	1424	278.3	909	317.3	1112	369
Min (μm^2)	10	10	13.3	12.1	24.8	22.5	22.5	22.5
No. of grains	133	459	91	2977	62	3840	256	3115

before [35]. All plotted compositions fall within the bcc regions of the phase diagrams except for Composition 2, which falls within the σ phase region. The compositions of Region C of the Ti−6Al−4V/V/SS304L FGM and Region 4 of the SS420/V FGM are also marked, in order to compare the three compositions at which the σ phase formed.

The simulated thermal history profiles determined using Netfabb Simulation for the SS420/V/Ti−6Al−4V and Ti−6Al−4V/V/SS304L FGMs are given in Fig. 10. From this, the approximate initial cooling rates for both FGMs were calculated, and it was found that

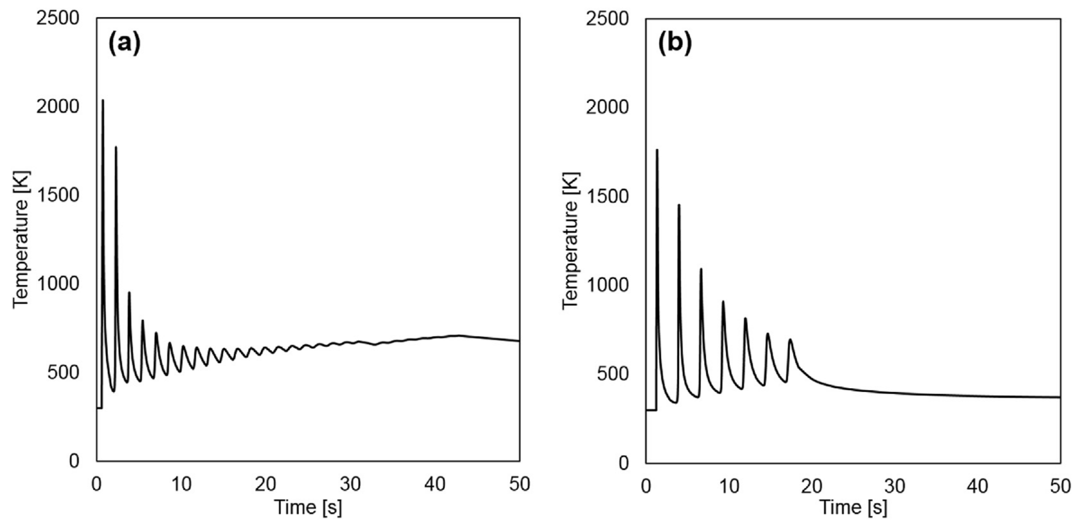

Fig. 10. Simulated temperature profiles during deposition, at a location of (a) the first layer of the planned 100 vol% V composition in the SS420/V/Ti–6Al–4V FGM and (b) the first layer of the planned 50 vol% SS304L/50 vol% V composition in the Ti–6Al–4V/V/SS304L FGM.

they were similar; the SS420/V/Ti–6Al–4V FGM had an initial cooling rate of approximately 5.1×10^3 K/s, while the cooling rate for the Ti–6Al–4V/V/SS304L FGM was calculated to be approximately 5.8×10^3 K/s.

In order to perform thermodynamic and kinetic simulations, the temperature range over which the σ phase is stable was determined. At the compositions corresponding to regions where the σ phase was detected, i.e., Fe-40.9V-11.8Cr-6.8Ni (wt. %) and Fe-39.6V-9.0Cr-1.9Ti (wt. %) in the Ti–6Al–4V/V/SS304L and SS420/V/Ti–6Al–4V FGMs, respectively, the σ phase becomes thermodynamically stable below 1153 K for the Ti–6Al–4V/V/SS304L FGM and 1264 K for the SS420/V/Ti–6Al–4V FGM. The increased stability of the σ phase at higher temperatures in the SS420–V alloy is attributed to the increased V to Cr ratio in the SS420–V alloy compared to SS304L–V alloy, because the binary (Fe,V)-σ phase is more stable at higher temperatures than the binary (Fe,Cr)-σ phase.

Fig. 11 shows the equilibrium phase fractions for the SS304L–V and SS420–V alloys. In both cases, the σ phase is stable at low temperatures, with only bcc being stable at high temperatures to the melting point. The TTT diagram in Fig. 12 shows the time- and temperature-dependent precipitation of the σ phase from a bcc matrix for the SS304L–V and SS420–V alloys overlaid with cooling rates determined by the FEA simulations. The TTT curves are similar in shape, however the SS420–V alloy is predicted to precipitate the σ phase at higher temperatures and shorter times than the SS304L–V alloy. The cooling curve corresponding to the first layer in the SS420–V alloy intersects with the bulk of the SS420–V alloy TTT curve, while the first SS304L–V alloy cooling curve briefly passes through the nose of the SS304L–V alloy TTT curve. The results of DICTRA simulations of the growth of the σ phase in SS304L–V and SS420–V alloys at 1100 K are shown in Fig. 6b. At this temperature, with total growth distance kept the same (50 nm) for both alloys, the growth of the σ phase in the SS420–V alloy is faster than in the SS304L–V alloy.

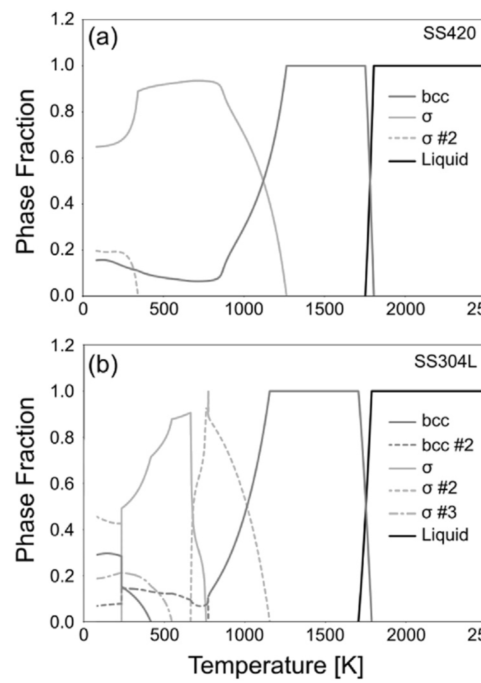

Fig. 11. Equilibrium phase fractions of the (a) SS420–V alloy in the σ-containing region and (b) the SS304 L–V alloy in the σ-containing region, with the number after the phase name denoting the composition sets, i.e., miscibility gaps. In the SS420/V alloy, the σ phase extends to higher temperatures than in the SS304 L/V alloy.

5. Discussion

5.1. Simulated nucleation and growth of the σ phase

In the studied SS304L–V and SS420–V alloy compositions, the σ phase is stable only at low temperatures. According to the equilibrium phase fractions, alloys cooling from the melt should solidify into a single-phase bcc alloy, which was observed in these FGMs in the regions surrounding the σ phase. Therefore, any σ phase that forms via the AM process must be through a solid-state reaction, nucleating from the bcc phase and growing throughout the remainder of a build. The TTT diagrams indicated that the σ phase should precipitate in the SS420–V alloy before the SS304L–V alloy for all time and temperature combinations. The Netfabb simulation predictions of the alloy cooling rates showed that it was more favorable for the σ phase to form in the SS420–V alloy compared to the SS304L–V alloy, given that the cooling curves are predicted to pass through the nose of the TTT diagrams. Together, the TTT curves and cooling rate simulations indicated that nucleation cannot

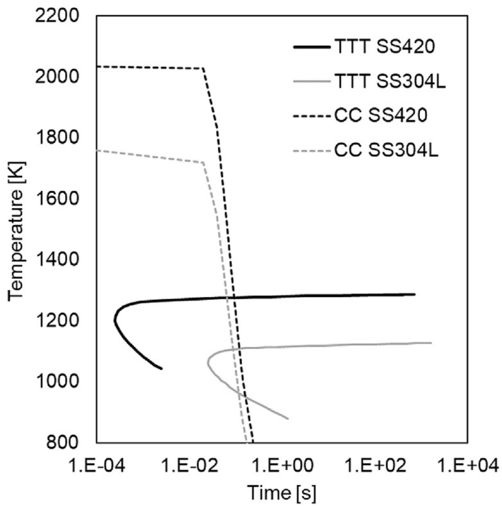

Fig. 12. Time-temperature-transformation (TTT) curves for precipitation of the σ phase from a bcc matrix calculated by TC-PRISMA. The black and gray solid lines depict the transformation curves for the SS420–V alloy in the σ region and the SS304L–V alloy in the σ region, respectively. The cooling curves (CC) determined from the initial cooling rate of the Netfabb simulations are overlaid on the TTT diagram.

account for the much higher amount of the σ phase in the Ti–6Al–4V/V/SS304L FGM compared to the SS420/V/Ti–6Al–4V FGM.

In the DICTRA simulations of the growth of the σ phase, it was assumed that the σ phase had already nucleated, therefore the growth rates are dominated by the diffusion of elements in the matrix to the σ phase interface. At 1100 K, the growth rate of the σ phase in the SS420–V alloy is an order of magnitude faster than that in the SS304L–V alloy. Therefore, under the similar cooling conditions predicted by the FEA simulations, the amount of σ phase growth in the SS420/V/Ti–6Al–4V FGM should be higher than that in the Ti–6Al–4V/V/SS304L FGM if diffusion-dominated growth was the sole factor controlling the amount of the σ phase. However, in the SS420/V/Ti–6Al–4V FGM, an additional 25 layers were added above the region containing the σ phase, while the Ti–6Al–4V/V/SS304L build was stopped after only six layers. This exposed the σ-containing region of the SS420/V/Ti–6Al–4V FGM to many more thermal cycles due to the additional layers, which would also favor more growth of the σ phase in the SS420/V/Ti–6Al–4V FGM compared to the Ti–6Al–4V/V/SS304L FGM. Since the SS420/V/Ti–6Al–4V FGM had significantly less σ phase, the growth of any nucleated σ phase does not account for the difference in the σ phase fraction.

Because both simulations predicted that the nucleation and growth of the σ phase should occur more readily in the SS420/V/Ti–6Al–4V FGM than the Ti–6Al–4V/V/SS304L FGM, we can conclude that nucleation and growth of the σ phase in these bcc alloys cannot explain the relatively large amount of the σ phase present in the Ti–6Al–4V/V/SS304L FGM compared to the SS420/V/Ti–6Al–4V FGM.

5.2. Mixing between layers

As described in Section 4.1.1, the SS420/V/Ti–6Al–4V FGM consisted of two deposited composition changes, but due to partial re-melting and mixing between layers, seven discrete regions with distinct chemistries were identified (Fig. 7). V only reaches a maximum composition of approximately 65 wt% in the Composition 3 region, where 100 vol% of V powder was deposited. There was also intermixing in the transition from the 100% deposited V composition and the 100% deposited Ti–6Al–4V composition, which is shown by the propagation of V content in Composition 4. These composition differences highlight the intermixing between layers that occurs during the additive manufacturing process in which previous layers are re-melted during the deposition of new layers in order to produce a metallurgical bond between layers. The mixing between the last layer of SS420 and the first layer of V is more significant due to the large increase in power. SS420 was deposited at a laser power of 600 W and V was deposited at a laser power of 800 W. This change in parameters was implemented due to the relatively high melting temperature of V (2183 K [36]) compared to the solidus/liquidus temperature of SS420 (1727/1783 K [37]). The increase in laser power creates a deeper melt pool in the layers directly preceding the change compared to the previous layers. This resulted in significant re-melting of the last layer of SS420 and mixing between the first layer of V and the last layer of SS420. Mixing between layers led to accessing compositions between the discrete planned steps that are favorable to the σ phase nucleation.

5.3. Crack formation and the σ phase growth

There are specific compositions that are favorable for the σ formation, as shown in the Fe–Cr–V isothermal section in Fig. 1. All three FGMs had gradient compositions that fell within this range: Composition 2 of the SS420/V/Ti–6Al–4V FGM, Region C in the Ti–6Al–4V/V/SS304L FGM, and Region 4 in the SS420/V FGM. Given their locations on the isothermal section, and it would be expected that the σ phase would form at each of these compositions.

As described in Sections 4 and 5.1, all three samples had varying amounts of the σ phase, as well as different types of cracking, and nucleation and growth were eliminated as reasons for the difference in the phase fraction of the σ phase among the three FGMs. The SS420/V/Ti–6Al–4V sample had the least amount of the σ phase, as determined by EBSD, in which the analyzed area covered approximately 6.9 mm^2 and only 0.41% of that area was identified as the σ phase. In this sample, cracks were present near the areas where the σ phase nucleated, but these cracks were small and did not preclude the finishing of the build. Larger cracks were present in the Ti–6Al–4V/V/SS304L and SS420/V FGMs. In the Ti–6Al–4V/V/SS304L FGM, a significant amount of the σ phase was present in Region C, specifically in the areas above the large cracks that spanned this region. Directly above the crack, the area analyzed with EBSD covered 0.03 mm^2 and 33% of that area was identified as the σ phase. Similarly, in the SS420/V FGM, the σ phase observed in Region 4 was concentrated in the area above the large cracks on the outer edge of the sample. These results show a correlation between the amount of σ phase that formed and the crack size and proximity of an area to the crack.

We observed that large cracks resulting in separation of material appear to have caused an interruption in the heat conduction path during the repeated heating and cooling cycles in the AM fabrication process. This resulted in inefficient heat conduction, and therefore, heat build-up in the area directly above the crack. The σ phase growth is favorable with increasing temperature, so when the material above the crack remains at higher temperatures for a longer period of time, the growth of the σ phase is promoted. This extended time spent at elevated temperatures is further evidenced by comparing the grain size in areas above the crack with areas below or adjacent to the crack, or at the center of the sample. As described in Section 4 and shown in Table 7, in the area above the crack in the SS420/V FGM, the average grain size of the σ phase is about six times larger than in other areas of the sample.

A schematic of the cooling curves associated with both types of

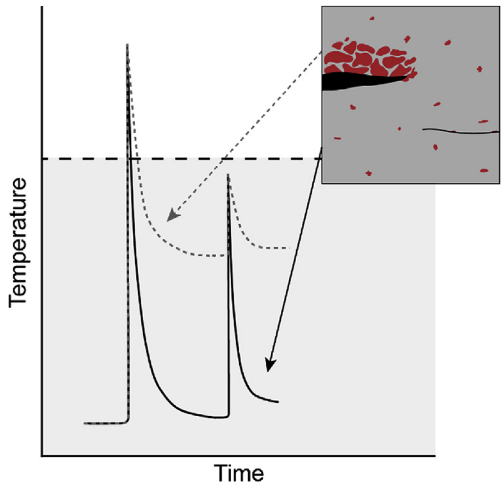

Fig. 13. Schematic illustrating that increased time at elevated temperatures results in growth of σ phase, with the dashed horizontal line representing the upper temperature limit for σ phase formation. The two heating and cooling profiles (one shown as solid lines, the other as dashed) represent the thermal profile at two different locations within the inset schematic of representative sample. The black curve represents an area with minimal cracking and minimal σ phase growth, and the gray dashed curve represents an area with large cracks and large amounts of σ phase. These differences indicate that the cracking is the cause of the growth σ phase rather than a consequence of the phase's growth.

σ phase growth is given in Fig. 13. This schematic illustrates the working hypothesis of the growth of σ phase in FGMs, in that an area with minimal σ phase growth, similar to the SS420/V/Ti–6Al–4V FGM, would spend minimal time at high temperatures, which is reflected in the black cooling curve, and which would be seen if the heat conduction pathway is not interrupted with a significant crack. However, an area with large amounts of σ phase growth, similar to the Ti–6Al–4V/V/SS304L and SS420/V FGMs, would have a cooling curve similar to that of the gray dashed line, which remains at high temperatures for longer amounts of time, perhaps due to the heat conduction pathway being interrupted by a significant crack.

Note, that fracture initiation and crack propagation are impacted by the build-up of residual stresses and thermal stresses during processing. These stresses are present as a result of the rapid solidification during the additive manufacturing fabrication process as well as the addition of layers on top of already solidified material. When disparate phases are present at a location in a component, and these phases have differences in stiffness, yield strength, or coefficient of thermal expansion, the residual or thermal stresses may be sufficient to drive fracture initiation at stress concentrations in the multi-phase regions. The samples in this study all had two-phase regions with bcc and the σ phase.

6. Summary and conclusion

In this study, the differing amounts of the σ phase in three FGMs made by DED AM, SS420 to V to Ti–6Al–4V, Ti–6Al–4V to V to SS304L, and SS420 to V, were compared and simulations were performed to determine the cause for different amounts of this phase at similar compositions within these samples. The elemental and phase compositions were experimentally determined and compared to thermodynamic and kinetic evaluations of the systems. The primary findings of this study are as follows:

- Of the three FGMs studied, almost no σ phase was found in the SS420/V/Ti–6Al–4V FGM, while notable amounts of σ phase were found above cracks in both the Ti–6Al–4V/V/SS304L and SS420/V FGMs.
- The TTT curves simulated using TC-PRISMA for the SS420–V and SS304L–V alloys show that the SS420–V alloy should nucleate the σ phase from the bcc matrix faster than the SS304L–V alloy.
- The DICTRA simulated growth of the σ phase in SS420–V and SS304L–V alloys shows that at 1100 K the σ phase in the SS420–V alloy is predicted to both start and finish growing before the SS304L–V, and grow to a higher σ phase fraction compared to the SS304L–V.
- EBSD analysis showed that the difference in the amount of the σ phase was dependent on location relative to cracks, rather than differences in chemistry, in the SS420/V and Ti–6Al–4V/V/SS304L FGMs. Within a single gradient composition region, the amount of the σ phase was highest above the crack and decreased further away from the crack.
- EBSD analysis showed that the SS420/V/Ti–6Al–4V FGM had minimal amounts of the σ phase near the microcracks in the intermixing region between the planned 100 vol% SS420 composition and the 100 vol% V composition.
- Given that the thermodynamic and kinetic simulation results differ from the experimental observations, the key difference in the amount of the σ phase was attributed to different thermal histories due to the different levels of cracking in each of the samples. It was determined that the σ phase growth in the Ti–6Al–4V/V/SS304L and SS420/V FGMs was a result of the build-up of heat above the crack due to the crack interrupting the heat conduction path rather than the cracking occurring as a result of σ phase growth. Smaller microcracks in the SS420/V/Ti–6Al–4V provided enough material contact for heat to dissipate through the sample, resulting in minimal amounts of the σ phase growth.
- This study highlights the importance of combining theoretical calculations and experimental characterization. If analyzing these FGMs independently, and only looking at experimental phase data, one might conclude that the cracks seen in the FGMs resulted because of the presence of σ phase, rather than recognizing that the cracks of varying severity were the reason for the disparate growth of σ phase in these FGMs.

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